Little's Formula

2006-10-03

Little's Formula

- Important quality of service issue is the average delay.
- We would also like to be able to compute
- the average number in the queue and
- the average time spent queueing.
- We introduce Little's formula, a very general formula.

Little's Formula (continued)

- Can be applied to $G/G/1$ queues.
- Gives the relationship between
- Steady-state average size of a queue, $\bar{N}$.
- The steady-state arrival rate, λ.
- The steady-state customer delay, $\bar{T}$.

Time and Steady State Averages

- We have already introduced the distribution means and sample based means.
- Time based averages are a further type of average.
- The time average is the average value of a particular function over time.

$$\bar{f}_t = \frac{1}{t} \int_0^t f(\tau)\, d\tau.$$

Time and Steady State Averages II

- The derivation of Little's formula relies on an assumption.
- Assume that averages under the steady state distribution are equivalent to the time average in the limit as $t \to \infty$,
- Average number in the system as a function of time is given by

$$\bar{N}_t = \frac{1}{t} \int_0^t N(\tau)\, d\tau$$

and the time average after in the limit as $t \to \infty$ is given by

$$\bar{N} = \lim_{t \to \infty} \bar{N}(t).$$

which we assume is equivalent to $\langle k \rangle_{p(k)} \equiv \bar{N}$. See pg 156-157 in (Bertsekas and Gallager, 1992).

The Formula

- Applies under steady state conditions.
- Under the steady state assumptions it states,

$$\bar{N} = \lambda \bar{T}.$$

The Formula (continued)

- **Example**: Nightclub queue. You are driving past a nightclub, wondering how long you will have to wait to go in. You notice on Thursday nights that the queue typically contains 20 people and that people arrive at a rate of 40 per hour. According to Little's formula, how long would you expect to have to wait to get in? **Answer**
- The waiting time is given by Little's formula: $\bar{N} = 20$, $\lambda = 40$ and $\bar{T} = \frac{\bar{N}}{\lambda} = \frac{20}{40} = \frac{1}{2}\text{hour}$.

Intuition

- If we see there are twenty people in a queue and they enter a building at a rate of 40 per hour, we might expect to have to wait half-an-hour in the queue. Formally though, we will build upon the intuition with a simple graphical 'proof' of the formula based upon that given in (Bertsekas and Gallager, 1992).

Graphical Proof I

- Example 1, Chapter 7. Graphical Proof of Little's Formula.

Graphical Proof II

- Sketch two curves vs time: the number of arrivals ($\alpha(\tau)$) and the number of departures ($\beta(\tau)$) .
- Also plot $N(t)$ the number in the queue as function of time.
- By looking at Figure **??** we note that the area between the lines in the top figure can be expressed as both $\int_0^t N(\tau)d\tau$ and $\sum_{i=1}^{\alpha(t)} T_i$. We may therefore equate these expressions,

$$\int_0^t N(\tau)\, d\tau = \sum_{i=1}^{\alpha(t)} T_i.$$

Little's Formula Derivation

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$$\frac{1}{t} \int_0^t N(\tau)\, d\tau = \frac{\alpha(t)}{t}\, \frac{\sum_{i=1}^{\alpha(t)} T_i}{\alpha(t)}.$$

The equation now has the form of three averages.

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Little's Formula

- $$\bar{N} = \lambda \bar{T}$$

Little's Formula (continued)

- Applies regardless of the arrival and service time distributions.
- Relies on the system being in *steady state*.
- Can be used at the level of the system.
- Or the level of the queue.
- Or the level of service.

Average Delay in $M/M/1$

- The mean of the steady state distribution for an $M/M/1$ queue is

$$\bar{N} = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho}.$$

- Can use Little's formula to tell us the average delay.
- From $\bar{N} = \lambda \bar{T}$ we rearrange to recover

$$\bar{T} = \frac{1}{\mu - \lambda}$$

Average Delay in $M/M/1$ (continued)

- This is not the time spent in the queue!
- This is the time spent in the system: it includes*service time.*
- arises from *propagation delay.*
- length of phone call *etc*.

Little's Formula for Queueing Time and Transmission Time

- The total delay can be divided into
- queueing time, W,

Little's Formula for Queueing Time and Transmission Time (continued)

- and service time X.

$$T = X + W.$$

- Similarly the average delay may be decomposed as

$$\bar{T} = \bar{X} + \bar{W}.$$

- The queue and transmission may the be analysed independently using Little's formula.

The Transmission

- Assume that customers/packets do not leave the queue.
- Rate of arrivals at the queue must equal the rate of arrivals at transmission.

The Transmission (continued)

- We define the the average number under transmission as ρ

$$\rho = \lambda \bar{X},$$

where $\bar{X}$ is the average time spend under transmission for each packet. * For the $M/M/1$ queue the average transmission time is given by $\bar{X} = \frac{1}{\mu}$, * So for the $M/M/1$ queue we have

$$\rho = \frac{\lambda}{\mu},$$

a formula which we have already used in the last chapter.

Utilisation Factor

- ρ is the utilisation factor.
- Represents the fraction of time that the transmission line (or server) is in use.

Utilisation Factor (continued)

- It is a measure of efficiency.
- It is also the steady state probability that the transmission line is in use.
- For a lossless system this probability is given by $1 - P(0)$.

Utilisation Example

- Example 2, Chapter 7. Simple example of Little's formula. You are at a queue in a bar and notice that the bar tender spends 10% of her time cleaning between customers. If the typical number of people waiting to be served is 3 what is the ratio of time spent serving to time being served?

Utilisation Example (continued)

The Queue

- For arrival rate λ, and $\bar{N}_Q$, the average number in queue, and $\bar{W}$, the average queueing (waiting) time.

$$\bar{N}_Q = \lambda \bar{W}$$

- For the $M/M/1$ queue

$$\bar{T} = \bar{X} + \bar{W} = \frac{1}{\mu} + \bar{W}$$

so the average time spent waiting in the queue is given by

$$\bar{W} = \bar{T} - \frac{1}{\mu} = \frac{\rho}{\mu - \lambda}.$$

- Average number in the queue is then

$$\bar{N}_Q = \frac{\rho^2}{}$$

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- Little's theorem and the $M/M/1$ delay formulae of Bertsekas and Gallager (1992)

Exercises

- Lab 1: sample-mean occupancy versus $\bar{N} = \rho/(1-\rho)$, and when the time average is not yet the steady state of Lawrence (2006)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.

Lawrence, N.D., 2006. Discrete simulation of the $M/M/1$ queue.